CONSERVATION PLAN

Whole farm - three tracts

Smith Family Farms, LLC
4820 Cross Keys Road
Harrisonburg, Rockingham County, VA 22801

FARM: 4417 TRACTS: 2208, 2211, 2214
ACC ID: 26051004417

TOTAL CROPLAND 408.0 ac TOTAL PASTURE 201.0 ac
ALL ACRES AND BOUNDARIES ARE APPROXIMATE

OBJECTIVE

To maintain a viable corn silage and small grain rotation while reducing nutrient and sediment delivery to Cub Run and Mill Creek, and to improve forage distribution for the cow-calf herd.

The producer intends to retire a portion of the cropland on the home tract into perennial pasture over the planning period.

RESOURCE CONCERNS

- Sheet and rill erosion on cropland slopes above both streams.
- Livestock have direct access to Cub Run in the lower pasture block.
- Nutrient management records incomplete for the east fields.
- Forage distribution limited by a single watering point.

PLAN CONTENTS

- Conservation plan map, soils map and resource concerns map, per tract
- Soils inventory report
- Practice schedule
- Grazing plan
- Operation and maintenance
- Verification schedule
- Programme and cost-share notes
- Signatures

CONSERVATION PLAN MAP

Tract 2208 - Home tract

Smith Family Farms, LLC
4820 Cross Keys Road
Harrisonburg, Rockingham County, VA 22801

FARM: 4417 TRACT: 2208 WATERSHED: 0207
ACC ID: 26051002208

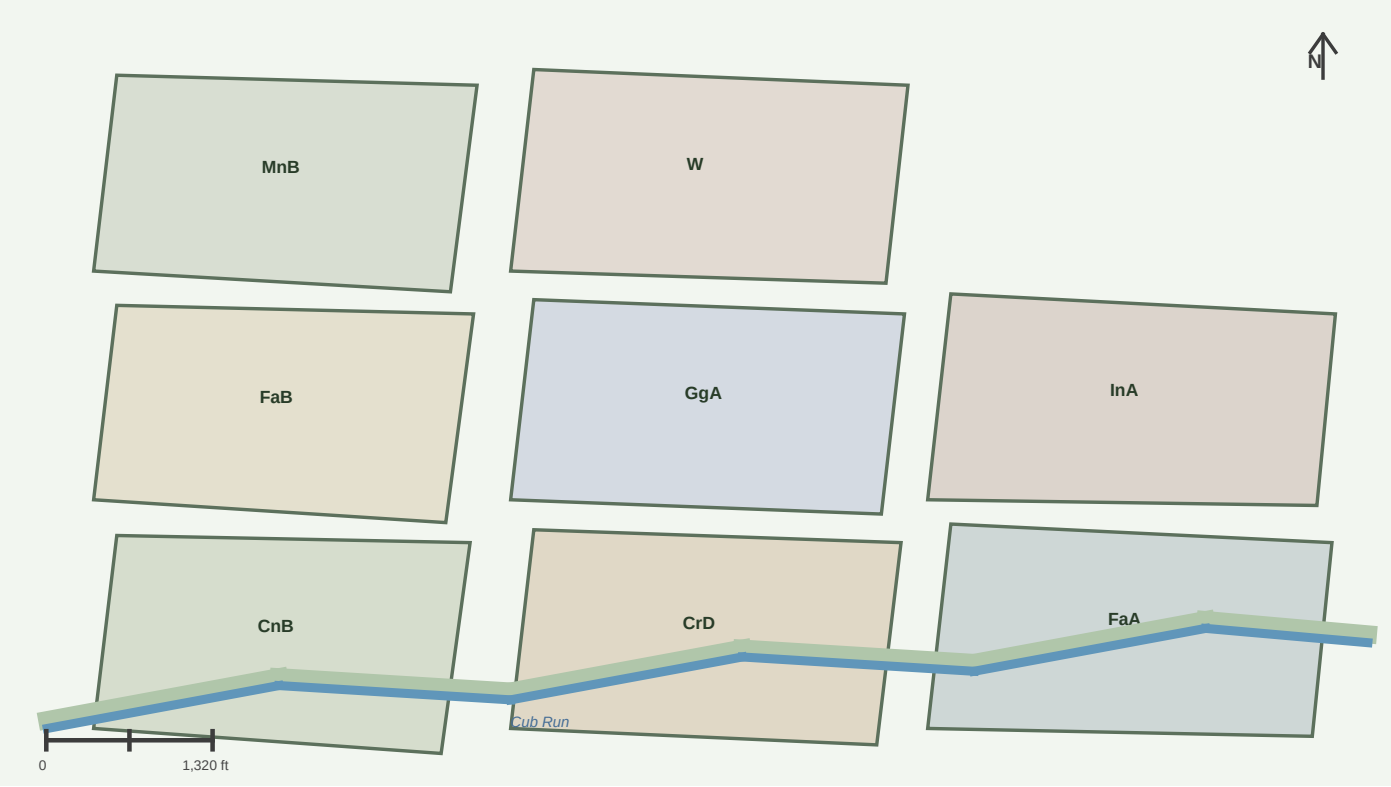
CROPLAND 214.0 ac PASTURE 96.0 ac WOODLAND 22.0 ac
CUB RUN FRONTAGE - 3,150 ft
ALL ACRES AND BOUNDARIES ARE APPROXIMATE



- Cropland
- Pasture
- Riparian buffer, planned or existing
- Barnyard / heavy use area
- Perennial stream - Cub Run

SOILS MAP

Tract 2208 - Home tract

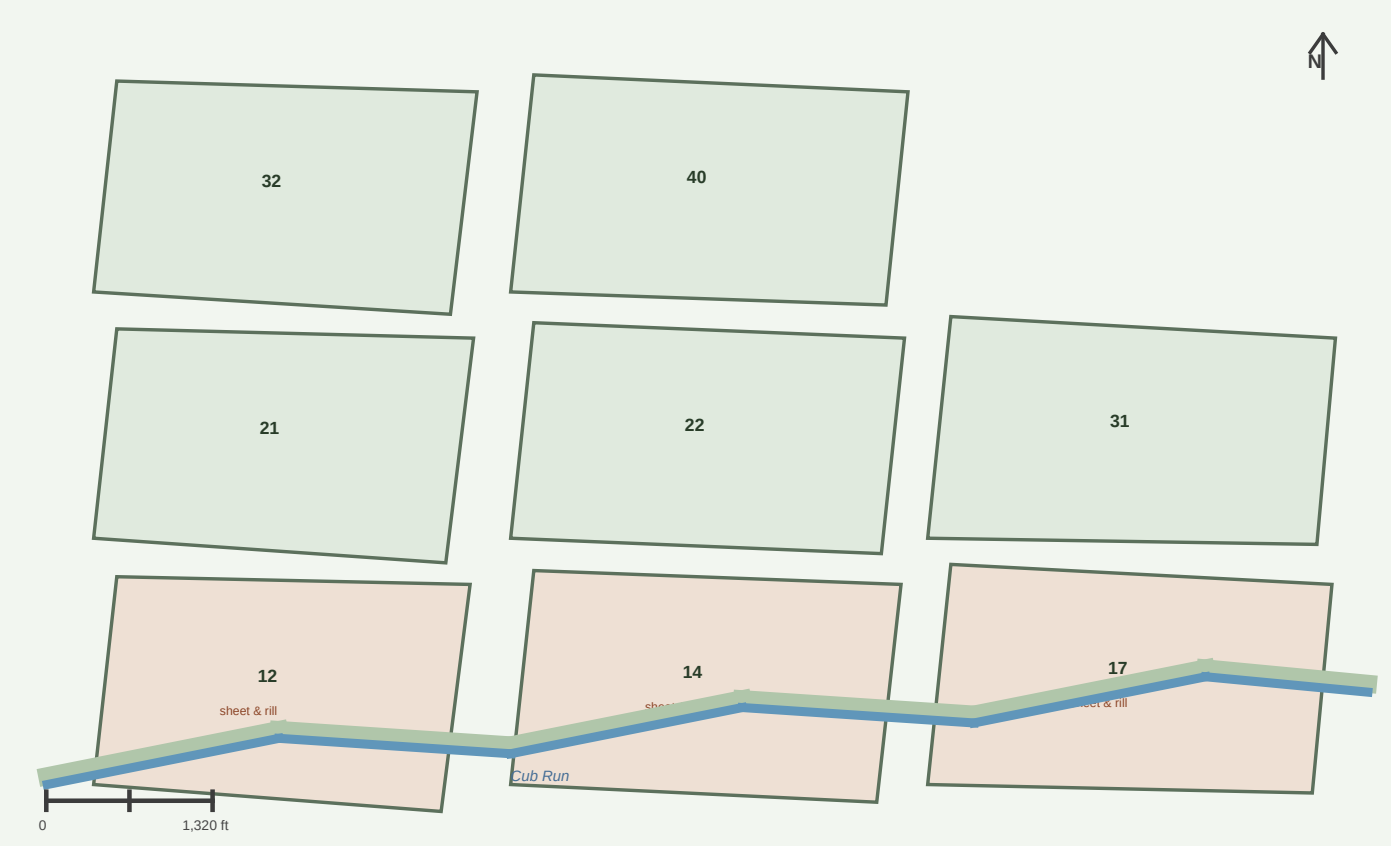


Soils Inventory Report

Symbol	Map Unit Name	Acres	Percent
CnB	Caneyville silt loam, 2 to 6 percent slopes	17.6	5%
CrD	Carbo-Rock outcrop complex, 15 to 25 percent slopes	10.7	3%
FaA	Frederick silt loam, 0 to 2 percent slopes	42.1	13%
FaB	Frederick silt loam, 2 to 7 percent slopes	35.8	11%
GgA	Guernsey silt loam, 0 to 2 percent slopes	50.3	15%
InA	Inman silt loam, 0 to 2 percent slopes	84.4	25%
MnB	Murrill loam, 2 to 7 percent slopes	58.3	18%
W	Water	2.3	1%
WeC	Weikert channery silt loam, 8 to 15 percent slopes	30.6	9%
Total:		**332.0**	**100%**

RESOURCE CONCERNS MAP

Tract 2208 - Home tract



- Sheet and rill erosion
- Forage distribution
- Direct livestock access to water

CONSERVATION PLAN MAP

Tract 2211 - Cross Keys tract

Smith Family Farms, LLC
4820 Cross Keys Road
Harrisonburg, Rockingham County, VA 22801

FARM: 4417 TRACT: 2211 WATERSHED: 0207
ACC ID: 26051002211

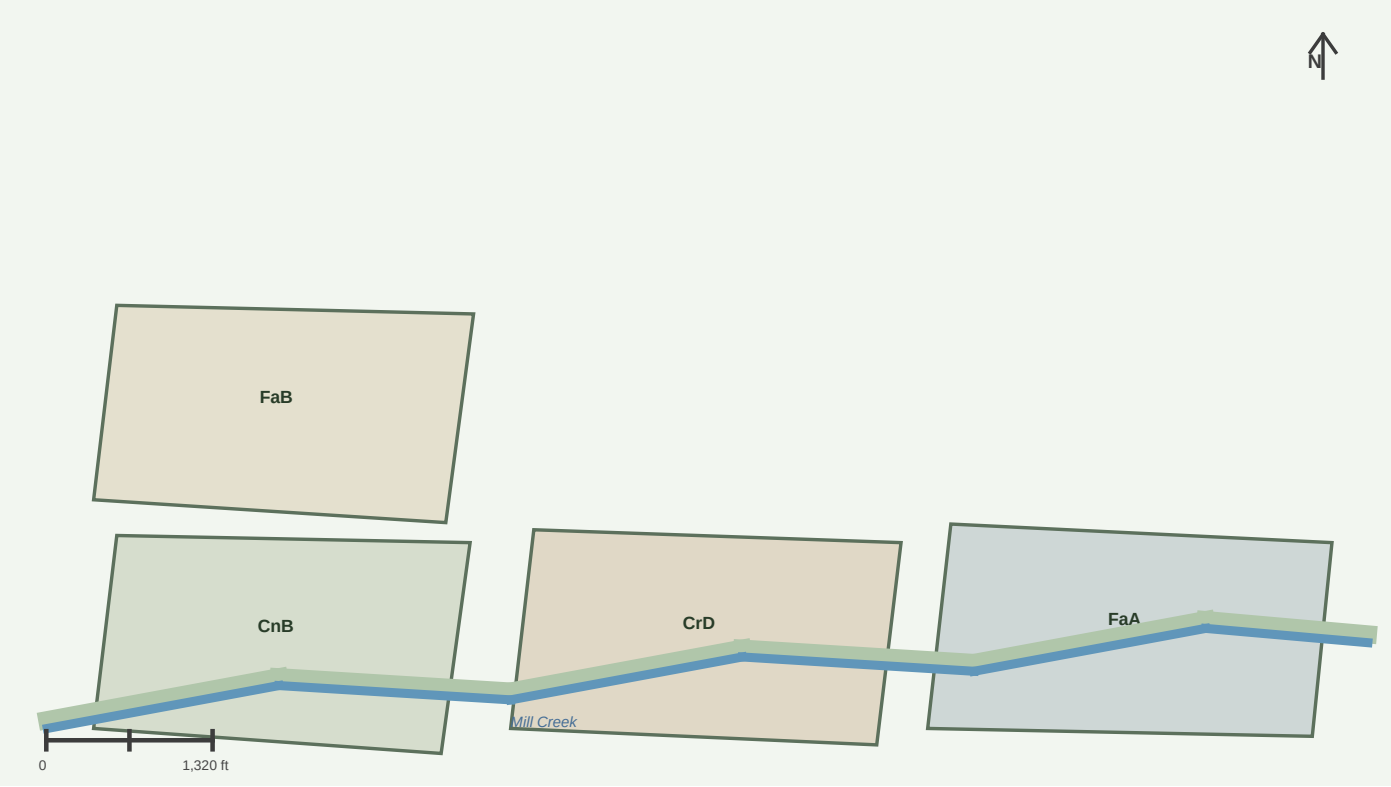
CROPLAND 118.0 ac PASTURE 44.0 ac WOODLAND 9.0 ac
MILL CREEK FRONTAGE - 1,820 ft
ALL ACRES AND BOUNDARIES ARE APPROXIMATE



- Cropland
- Pasture
- Riparian buffer, planned or existing
- Barnyard / heavy use area
- Perennial stream - Mill Creek

SOILS MAP

Tract 2211 - Cross Keys tract

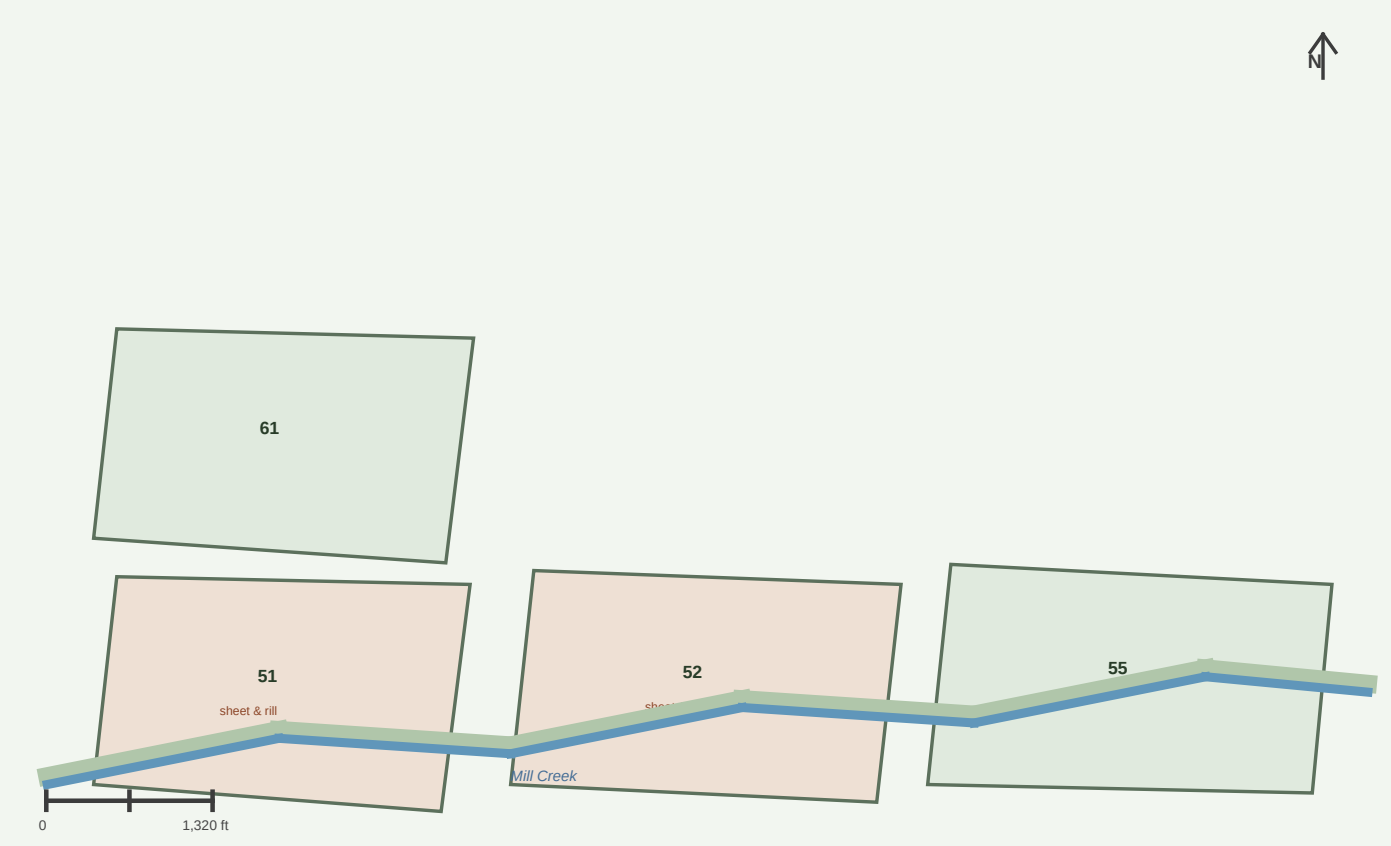


Soils Inventory Report

Symbol	Map Unit Name	Acres	Percent
CnB	Caneyville silt loam, 2 to 6 percent slopes	9.1	5%
CrD	Carbo-Rock outcrop complex, 15 to 25 percent slopes	5.5	3%
FaA	Frederick silt loam, 0 to 2 percent slopes	21.7	13%
FaB	Frederick silt loam, 2 to 7 percent slopes	18.5	11%
GgA	Guernsey silt loam, 0 to 2 percent slopes	25.9	15%
InA	Inman silt loam, 0 to 2 percent slopes	43.5	25%
MnB	Murrill loam, 2 to 7 percent slopes	30.0	18%
W	Water	1.2	1%
WeC	Weikert channery silt loam, 8 to 15 percent slopes	15.7	9%
Total:		**171.0**	**100%**

RESOURCE CONCERNS MAP

Tract 2211 - Cross Keys tract



- Sheet and rill erosion
- Forage distribution
- Direct livestock access to water

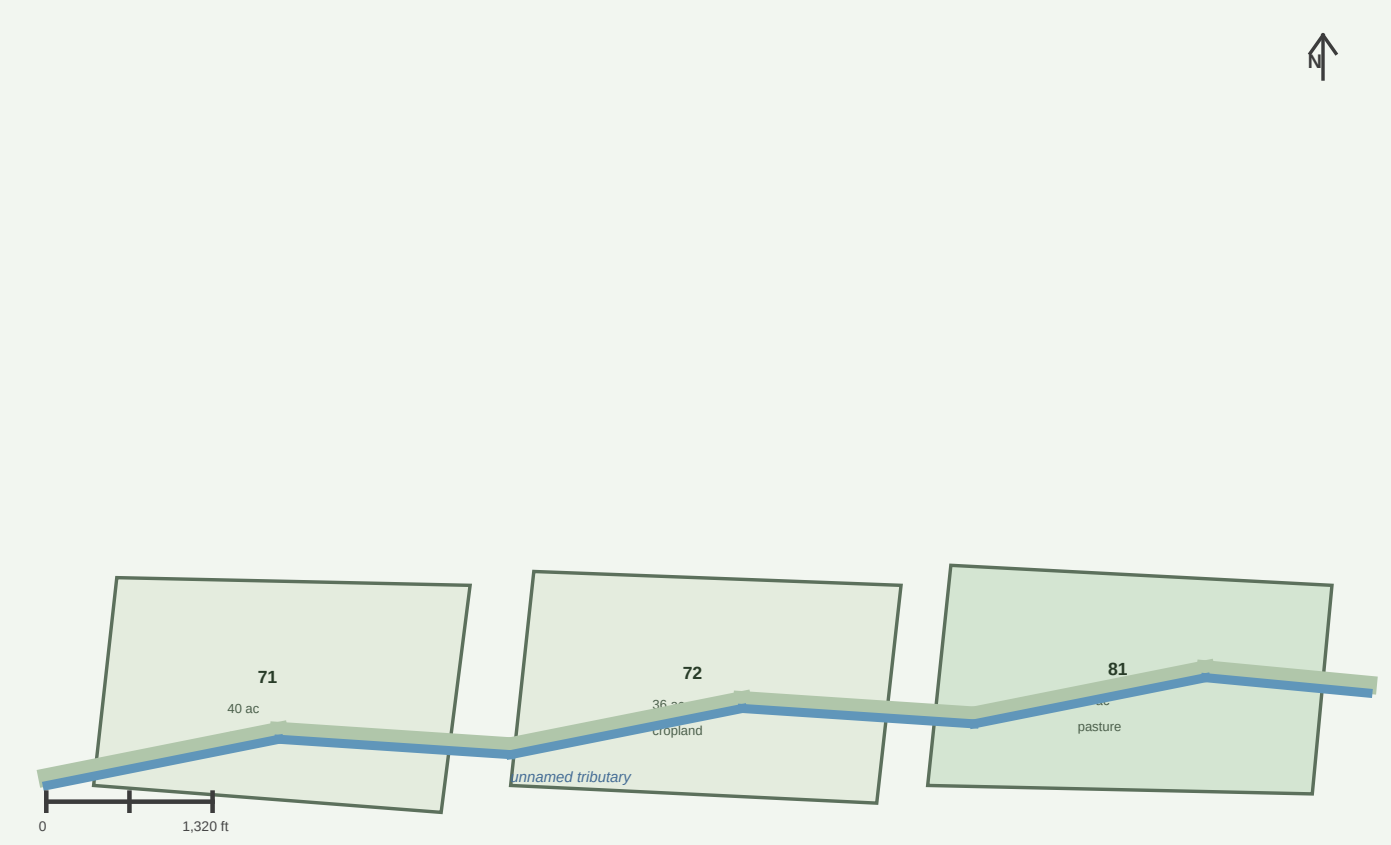
CONSERVATION PLAN MAP

Tract 2214 - Ridge tract

Smith Family Farms, LLC
4820 Cross Keys Road
Harrisonburg, Rockingham County, VA 22801

FARM: 4417 TRACT: 2214 WATERSHED: 0207
ACC ID: 26051002214

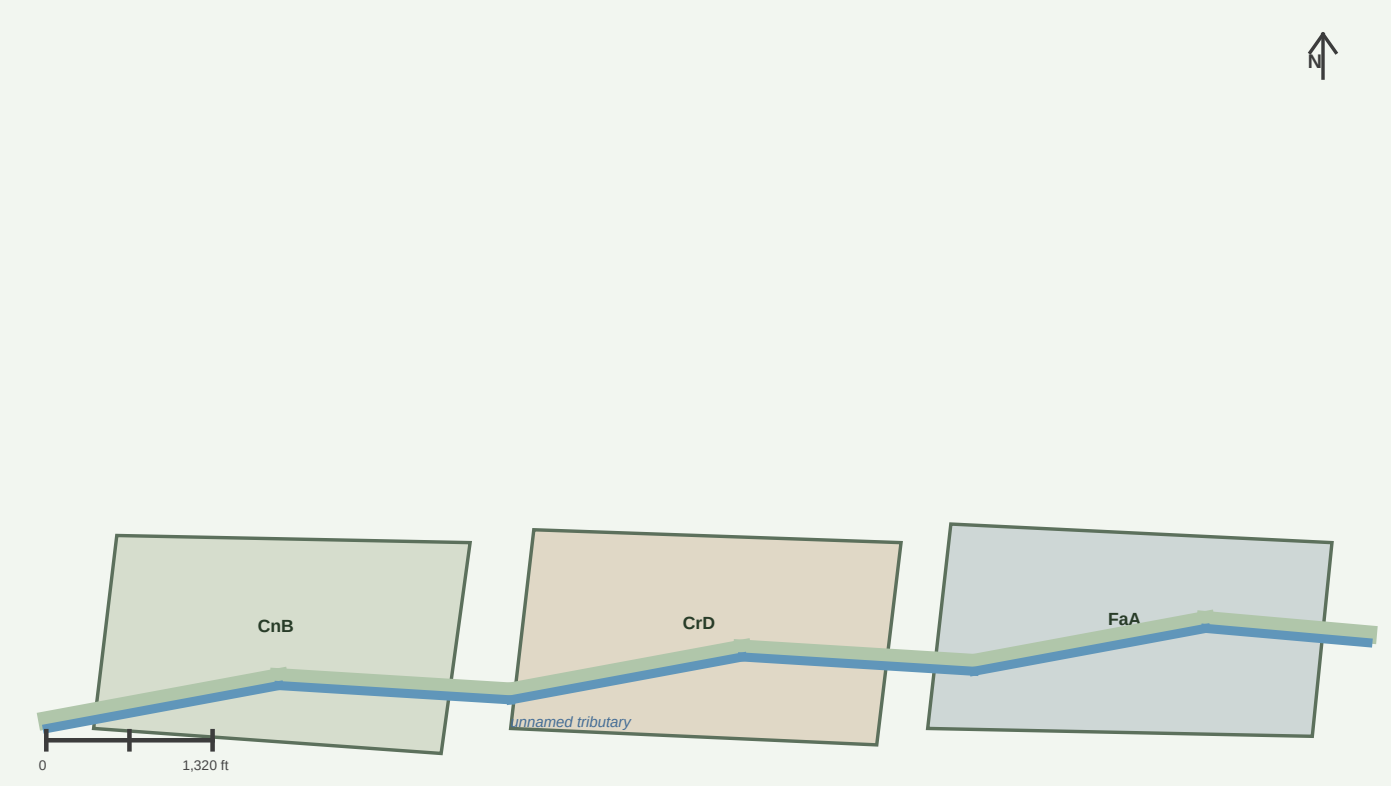
CROPLAND 76.0 ac PASTURE 61.0 ac WOODLAND 31.0 ac
UNNAMED TRIBUTARY FRONTAGE - 940 ft
ALL ACRES AND BOUNDARIES ARE APPROXIMATE



- Cropland
- Pasture
- Riparian buffer, planned or existing
- Barnyard / heavy use area
- Perennial stream - unnamed tributary

SOILS MAP

Tract 2214 - Ridge tract

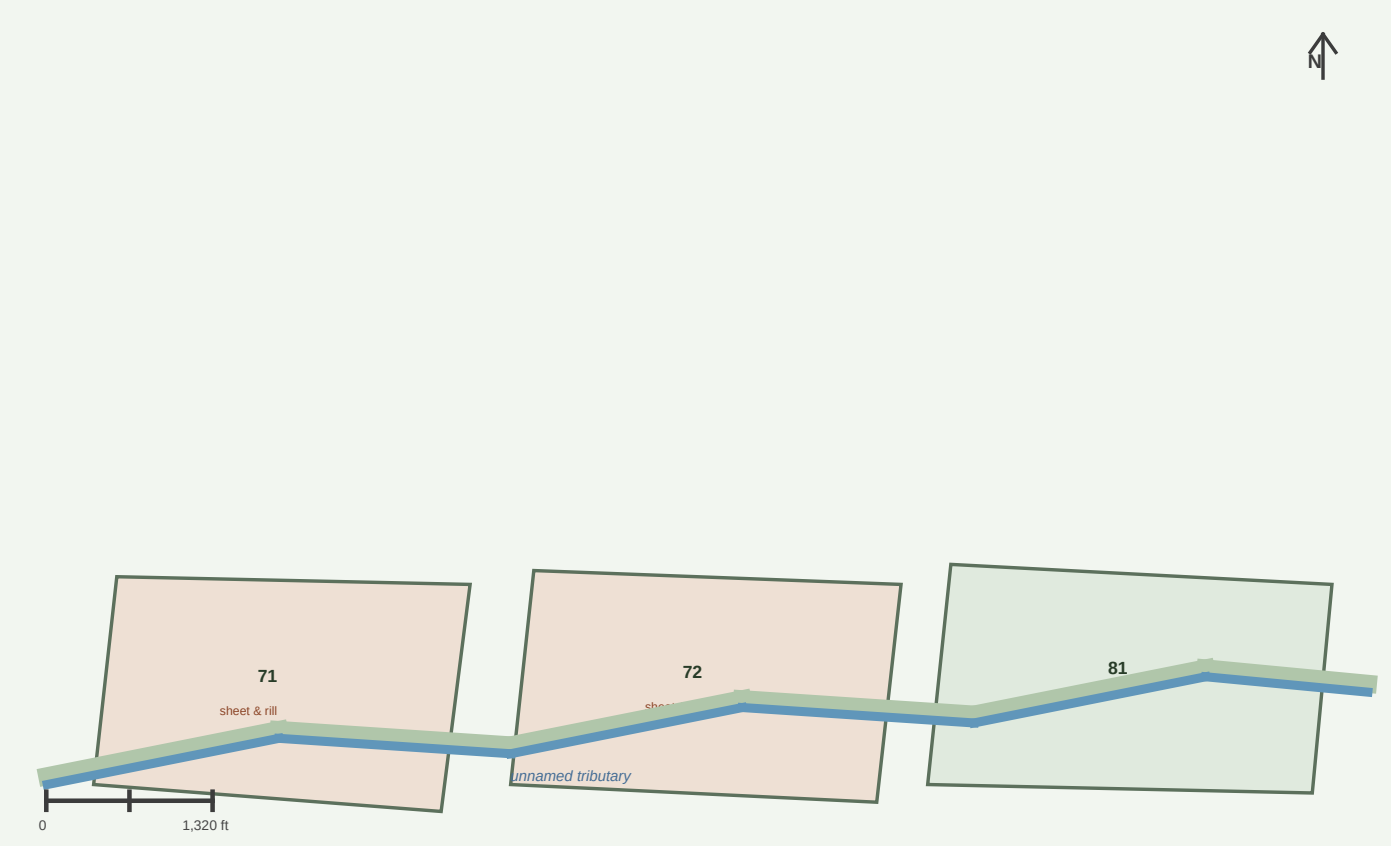


Soils Inventory Report

Symbol	Map Unit Name	Acres	Percent
CnB	Caneyville silt loam, 2 to 6 percent slopes	8.9	5%
CrD	Carbo-Rock outcrop complex, 15 to 25 percent slopes	5.4	3%
FaA	Frederick silt loam, 0 to 2 percent slopes	21.3	13%
FaB	Frederick silt loam, 2 to 7 percent slopes	18.1	11%
GgA	Guernsey silt loam, 0 to 2 percent slopes	25.4	15%
InA	Inman silt loam, 0 to 2 percent slopes	42.7	25%
MnB	Murrill loam, 2 to 7 percent slopes	29.5	18%
W	Water	1.2	1%
WeC	Weikert channery silt loam, 8 to 15 percent slopes	15.5	9%
Total:		**168.0**	**100%**

RESOURCE CONCERNS MAP

Tract 2214 - Ridge tract



- Sheet and rill erosion
- Forage distribution
- Direct livestock access to water

CONSERVATION PLAN

Practice schedule

Associated Ag Land Tracts: 2208, 2211, 2214

Cover Crop - 340
Cereal rye drilled following corn silage harvest. Not harvested. No fall nutrient application planned on any field in this schedule.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
12	64.0 ac	9	2026		
14	49.5 ac	9	2026		
17	38.0 ac	9	2026		
51	62.0 ac	9	2026		
52	56.0 ac	9	2026		

Total: 269.5 ac

Residue and Tillage Management, No Till - 329
Continuous no-till on all cropped acres. In place since 2019. Cost-shared under EQIP contract 68-3A75-19-112, which expired in 2024. No cost-share is recorded against the current compliance year.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
12	64.0 ac	4	2019	64.0 ac	5/2019
14	49.5 ac	4	2019	49.5 ac	5/2019
17	38.0 ac	4	2019	38.0 ac	5/2019
51	62.0 ac	4	2020	62.0 ac	4/2020

Total: 213.5 ac

Nutrient Management - 590
Whole-farm plan written 2024 by the district. Reductions for the 2025 reporting year were submitted to Virginia DEQ.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
ALL	408.0 ac	3	2024	408.0 ac	3/2024

Total: 408.0 ac

Riparian Forest Buffer - 391
Establish a 75-foot average width hardwood buffer along the west bank of Cub Run where the cropped field edge currently abuts the channel. Not yet installed.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
21	5.4 ac	11	2026		
22	3.1 ac	11	2026		

Total: 8.5 ac

Riparian Forest Buffer - 391
Existing 40-foot buffer on the east bank, planted 2016 under CREP. Retained and maintained. No further work planned.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
23	4.2 ac	3	2016	4.2 ac	4/2016

Total: 4.2 ac

Riparian Forest Buffer - 391
New 60-foot buffer on the Cross Keys tract along Mill Creek.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
55	4.2 ac	11	2027		

CONSERVATION PLAN

continued

Total: 4.2 ac

Prescribed Grazing - 528
Subdivide the lower pasture into six paddocks and rotate the cow-calf herd on a flexible schedule. No fixed move interval is specified; the producer will adjust to forage conditions.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
31	58.0 ac	5	2027		
32	38.0 ac	5	2027		
61	44.0 ac	5	2027		
81	61.0 ac	5	2028		

Total: 201.0 ac

Fence - 382
Exclusion fencing along both banks of Cub Run and Mill Creek, and perimeter fencing for the new paddock divisions.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
31	4,200 ft	5	2027		
61	2,600 ft	5	2027		

Total: 6,800.0 ft

Watering Facility - 614
Frost-free troughs with buried supply line from the existing well to serve the new paddocks. Nearest well is 900 feet from the paddock centre.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
31	2 no	5	2027		
61	2 no	5	2027		

Total: 4.0 no

Pipeline - 516
Buried waterline from the wellhead to the trough locations.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
31	1,850 ft	5	2027		

Total: 1,850.0 ft

Heavy Use Area Protection - 561
Concrete pad and roof runoff control at the existing feeding area adjacent to the barnyard.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
40	0.8 ac	8	2027		

Total: 0.8 ac

Conservation Crop Rotation - 328
Move fields 12 and 14 from continuous corn silage into a corn / small grain / soybean rotation.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
12	64.0 ac	4	2027		
14	49.5 ac	4	2027		

Total: 113.5 ac

Tree/Shrub Establishment - 612
Shade planting in the lower paddocks at the producer's request. Not a

CONSERVATION PLAN

continued

silvopasture establishment; confirm the applicable practice definition before claiming any reduction.

	Planned			Applied	
Field	Amount	Month	Year	Amount	Date
32	2.0 ac	11	2027		
			Total: 2.0 ac		

GRAZING PLAN

Prescribed Grazing - 528

OBJECTIVE
Improve forage utilisation and distribute grazing pressure away from the stream corridor on tracts 2208, 2211 and 2214.

PADDOCK LAYOUT

Field 31	58.0 ac	six paddocks, nominal 9.7 ac each
Field 32	38.0 ac	four paddocks, nominal 9.5 ac each
Field 61	44.0 ac	four paddocks, nominal 11.0 ac each
Field 81	61.0 ac	layout to be determined, year 3

MOVE SCHEDULE
No fixed interval is prescribed. The producer will move stock on forage condition, with a target residual height of 3 to 4 inches and a minimum 21 day rest in the growing season.

A written move schedule is NOT a condition of this plan. Where NRCS prescribed-grazing cost-share is sought, that programme's own plan requirement applies.

WATER
Two frost-free troughs per block, supplied from the existing well by buried pipeline. Nearest well is 900 ft from the paddock centre on tract 2208. Well yield to be confirmed before pipeline design.

FORAGE
Existing stand is predominantly tall fescue with mixed clover. Overseeding is not planned in the first two years.

SHADE
Producer has requested shade in the lower paddocks. Tree and shrub establishment is scheduled on field 32. This is a shade planting and is not proposed as silvopasture establishment.

OPERATION AND MAINTENANCE

COVER CROP - 340
Seed at the rate and date in the specification sheet. Do not apply fall nutrients to any field in this schedule. Terminate ahead of the following crop per label.

RESIDUE AND TILLAGE MANAGEMENT, NO TILL - 329
Maintain 15 percent or greater residue cover immediately after planting. Document residue counts annually.

RIPARIAN FOREST BUFFER - 391
Protect from livestock for the life of the practice. Replace losses to maintain the planted stem count through year three. Control competing vegetation for the first two growing seasons.

FENCE - 382
Inspect annually and after major storm events. Repair promptly where exclusion fencing is breached.

WATERING FACILITY - 614 / PIPELINE - 516
Drain and winterise as required. Inspect the supply line for leaks at each seasonal start-up.

HEAVY USE AREA PROTECTION - 561
Scrape and remove accumulated material as needed. Maintain roof runoff conveyance clear of debris.

VERIFICATION SCHEDULE

Virginia verifies on a five-year interval.

Practice	Interval	Responsible
Cover Crop - 340	annual	producer records
No Till - 329	annual	producer records
Nutrient Management - 590	annual	district
Riparian Forest Buffer - 391	5 year	district
Prescribed Grazing - 528	5 year	district
Heavy Use Area Protection - 561	5 year	district
Conservation Crop Rotation - 328	annual	producer records

Responsibility for the 2027 and later inspections has not yet been assigned. A staffed verification plan is required before the first structural practice is certified.

NOTE ON CREDIT DURATION

Credit duration is the period a practice counts WITHOUT inspection. With inspection it continues to be credited. A missed inspection is how a project silently loses its reductions.

PROGRAMME AND COST-SHARE NOTES

EQIP
Contract 68-3A75-19-112 covered no-till establishment on tracts 2208 and 2211. The contract expired in 2024. No cost-share is recorded against the current compliance year.

CREP
The 40-foot buffer on the east bank of Cub Run (field 23) was planted in 2016 under CREP and remains in place.

STATE REPORTING
Nutrient management reductions for the 2025 reporting year were submitted to Virginia DEQ.

VIRGINIA AGRICULTURAL COST SHARE
Perimeter and exclusion fencing, watering facilities and pipeline are expected to be cost-shared in part. Final shares to be set at contracting.

ITEMS THAT ARE COSTS, NOT PRACTICES
Fencing, pipeline and well work are necessary to deliver the grazing system. They are not themselves creditable reduction practices.

SIGNATURES

Producer

signature date

District Conservationist

signature date

District Representative

signature date

This plan is a synthetic document created for software testing. No real farm, producer or property is described. Field geometry, soils, acreages and practice schedules are invented.

PLAN SUMMARY

Blue Ridge Agricultural Services, LLC | PO Box 318, Bridgewater, VA 22812

This conservation plan covers three tracts totalling 609 acres in Rockingham County, Virginia. The operation is a corn silage and small grain rotation with a cow-calf enterprise on permanent pasture.

The producer's stated objectives are to reduce nutrient and sediment delivery to Cub Run and Mill Creek, to improve forage distribution, and to retire a portion of the cropland on the home tract into perennial pasture over the planning period.

Planning was carried out over three site visits between November 2025 and February 2026. Field boundaries were walked with the producer and checked against the most recent imagery.

EXISTING CONDITIONS

Blue Ridge Agricultural Services, LLC | PO Box 318, Bridgewater, VA 22812

Cropland is in continuous no-till, established 2019 under an EQIP contract that has since expired. Residue cover was measured at 38 to 52 percent after planting on the fields inspected.

Pasture is predominantly tall fescue with mixed clover, grazed continuously. Stand condition is fair. Forage utilisation is uneven: heavy near the single watering point and light at the field extremities.

Livestock currently have direct access to Cub Run in the lower pasture block on tract 2208. Streambank condition at the access points is poor, with active erosion over an estimated 180 linear feet.

A 40-foot riparian buffer was planted on the east bank in 2016 under CREP and is established. The west bank is unbuffered: the cropped field edge abuts the channel over roughly 1,400 feet.

RESOURCE CONCERNS IDENTIFIED

Blue Ridge Agricultural Services, LLC | PO Box 318, Bridgewater, VA 22812

SOIL EROSION - Sheet and rill
Cropland slopes above both streams. RUSLE2 output is attached. Current soil loss is within T on the no-till fields but the delivery ratio to the channel is high where no buffer exists.

WATER QUALITY DEGRADATION - Nutrients transported to surface water
Nutrient management plan written 2024. Records incomplete for the east fields in the 2025 crop year.

WATER QUALITY DEGRADATION - Pathogens and chemicals from manure
Direct livestock access to Cub Run.

INADEQUATE LIVESTOCK WATER - Quantity and distribution
Single watering point on 96 acres of pasture.

PLANT PRODUCTIVITY AND HEALTH - Inadequate structure and composition
Continuous grazing has reduced stand vigour in the lower paddocks.

PLANNED TREATMENT

Blue Ridge Agricultural Services, LLC | PO Box 318, Bridgewater, VA 22812

The treatment sequence is staged over three years.

YEAR 1 (2026)
Cover crop on all cropped acres. Establish the west bank riparian forest buffer on tract 2208.

YEAR 2 (2027)
Subdivide the lower pasture and install watering facilities and pipeline. Begin prescribed grazing. Install heavy use area protection at the barnyard. Begin the conservation crop rotation on fields 12 and 14. Establish the Mill Creek buffer on tract 2211.

YEAR 3 (2028)
Extend prescribed grazing to tract 2214. Review stand condition and determine whether overseeding is required.

ALTERNATIVES CONSIDERED

Blue Ridge Agricultural Services, LLC | PO Box 318, Bridgewater, VA 22812

ALTERNATIVE 1 - No action
Erosion and nutrient delivery continue at current rates. Livestock access to the stream is unaddressed. Rejected.

ALTERNATIVE 2 - Buffers and stream exclusion only
Addresses the stream directly at lower cost. Does not address forage distribution, and the producer indicated that fencing without a water solution would not be workable. Rejected as insufficient.

ALTERNATIVE 3 - Full treatment as planned (selected)
Buffers, exclusion fencing, water development and prescribed grazing together. Higher cost, addresses all identified concerns.

ALTERNATIVE 4 - Whole-farm conversion to pasture
Discussed. The producer is not willing to retire the full cropland base. A partial retirement is reflected in the crop rotation change.

PRODUCER DECISIONS AND NOTES

Blue Ridge Agricultural Services, LLC | PO Box 318, Bridgewater, VA 22812

The producer has agreed to the practice schedule as written, subject to cost-share availability for the fencing, pipeline and watering facilities.

The producer asked whether shade could be established in the lower paddocks. Tree and shrub establishment is scheduled on field 32. The producer was advised that this is a shade planting and should not be represented as silvopasture establishment.

The producer asked about the timing of any nutrient credit sale. He was advised that reductions already reported to the Commonwealth cannot be offered again, and that practices funded by a cost-share programme in a given year are not available for that year.

No written grazing move schedule is required by this plan. Where NRCS prescribed-grazing cost-share is sought, that programme's own plan requirement will apply.

MAP UNIT DESCRIPTION

Brief, generated

Generated brief soil descriptions are created for major soil components. The soil component is a part of the map unit. Each map unit is made up of one or more components.

This report is not a substitute for a site specific investigation.

Components in this survey area are listed on the following pages in order of extent within the planning unit.

COMPONENT: Frederick

Map unit symbol FaA, FaB - 31% of planning unit

The Frederick component makes up 85 percent of the map unit. Slopes are 0 to 7 percent. This component is on ridges on uplands. The parent material consists of residuum weathered from dolomite and limestone. Depth to a root restrictive layer is greater than 80 inches.

The natural drainage class is well drained. Water movement in the most restrictive layer is moderately high. Available water to a depth of 60 inches is about 9.2 inches. Shrink-swell potential is low.

This soil is not flooded and is not ponded. There is no zone of water saturation within a depth of 72 inches. Organic matter content in the surface horizon is about 2 percent. Nonirrigated land capability classification is 2e.

INTERPRETATIONS

Hydrologic soil group	B
Farmland classification	Prime farmland
Ecological site	F147XY231VA
Hydric rating	No

COMPONENT: Guernsey

Map unit symbol GgA - 21% of planning unit

The Guernsey component makes up 80 percent of the map unit. Slopes are 0 to 2 percent. This component is on hillslopes on uplands. The parent material consists of residuum weathered from interbedded shale, siltstone and limestone.

The natural drainage class is moderately well drained. Water movement in the most restrictive layer is moderately low. Available water to a depth of 60 inches is about 7.8 inches.

A seasonal zone of water saturation is at 18 inches during January through April. Nonirrigated land capability classification is 2w.

INTERPRETATIONS

Hydrologic soil group	B
Farmland classification	Prime farmland
Ecological site	F147XY221VA
Hydric rating	No

COMPONENT: Inman

Map unit symbol InA - 35% of planning unit

The Inman component makes up 90 percent of the map unit. Slopes are 0 to 2 percent. This component is on footslopes. The parent material consists of colluvium derived from shale and siltstone.

The natural drainage class is somewhat poorly drained. Water movement in the most restrictive layer is low. Available water to a depth of 60 inches is about 6.4 inches.

A seasonal zone of water saturation is at 12 inches during December through April. Nonirrigated land capability classification is 3w.

INTERPRETATIONS

Hydrologic soil group	B
Farmland classification	Prime farmland
Ecological site	F147XY235VA
Hydric rating	Partially hydric

COMPONENT: Murrill

Map unit symbol MnB - 24% of planning unit

The Murrill component makes up 88 percent of the map unit. Slopes are 2 to 7 percent. This component is on benches on uplands. The parent material consists of colluvium over residuum weathered from sandstone and limestone.

The natural drainage class is well drained. Available water to a depth of 60 inches is about 10.1 inches. Nonirrigated land capability classification is 2e.

INTERPRETATIONS

Hydrologic soil group	B
Farmland classification	Prime farmland
Ecological site	F147XY224VA
Hydric rating	No

COMPONENT: Weikert

Map unit symbol WeC - 13% of planning unit

The Weikert component makes up 92 percent of the map unit. Slopes are 8 to 15 percent. This component is on hillslopes. The parent material consists of residuum weathered from shale and siltstone.

Depth to a root restrictive layer, bedrock lithic, is 10 to 20 inches. The natural drainage class is somewhat excessively drained. Available water to a depth of 60 inches is about 2.1 inches, which is low.

Nonirrigated land capability classification is 6e. This component is not well suited to cultivated crops.

INTERPRETATIONS

Hydrologic soil group	C
Farmland classification	Farmland of statewide importance
Ecological site	F147XY213VA
Hydric rating	No

COMPONENT: Caneyville

Map unit symbol CnB - 7% of planning unit

The Caneyville component makes up 85 percent of the map unit. Slopes are 2 to 6 percent. The parent material consists of residuum weathered from limestone.

Depth to a root restrictive layer, bedrock lithic, is 20 to 40 inches. The natural drainage class is well drained. Available water to a depth of 60 inches is about 4.8 inches.

INTERPRETATIONS

Hydrologic soil group	C
Farmland classification	Farmland of statewide importance
Ecological site	F147XY207VA
Hydric rating	No

COMPONENT: Carbo-Rock outcrop

Map unit symbol CrD - 4% of planning unit

The Carbo component makes up 60 percent of the map unit and rock outcrop makes up 25 percent. Slopes are 15 to 25 percent.

Depth to a root restrictive layer, bedrock lithic, is 20 to 40 inches. Nonirrigated land capability classification is 7e. This map unit is not suited to cultivated crops and is best left in permanent cover.

INTERPRETATIONS

Hydrologic soil group	C
Farmland classification	Farmland of statewide importance
Ecological site	F147XY204VA
Hydric rating	No

CONSERVATION PRACTICE STANDARD

COVER CROP (340)

Code 340 Unit: Ac.

DEFINITION

Grasses, legumes, and forbs planted for seasonal vegetative cover.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Reduce erosion from wind and water.
- Capture and recycle or redistribute nutrients in the soil profile.
- Improve soil health and increase soil organic matter content.
- Reduce water quality degradation by utilizing excessive soil nutrients.
- Suppress weeds and break pest cycles.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

COVER CROP (340) - criteria

CRITERIA

Select species adapted to the site, planting date and intended purpose.
Plant at a rate and depth appropriate to the species and method.
Where nutrient capture is a purpose, establish the cover early enough to take up residual nitrogen before leaching losses occur.
Terminate at a time and by a method that does not compromise the following crop and that retains the required surface cover.
Do not harvest where the stated purpose is nutrient capture and the practice is being credited on that basis.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

Field Office Technical Guide, Section IV.
Virginia Agricultural BMP Manual, current edition.

CONSERVATION PRACTICE STANDARD

RESIDUE AND TILLAGE MANAGEMENT, NO TILL (329)

Code 329 Unit: Ac.

DEFINITION

Limiting soil disturbance to manage the amount, orientation and distribution of crop and plant residue on the soil surface year round.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Reduce sheet, rill and wind erosion.
- Improve soil health and increase soil organic matter.
- Reduce energy use.
- Increase plant-available moisture.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

RESIDUE AND TILLAGE MANAGEMENT, NO TILL (329) - criteria

CRITERIA

The full width of the soil surface shall not be disturbed.
Soil disturbance shall be limited to that caused by the planting or seeding operation, by nutrient injection, or by a strip no wider than one third of the row spacing.
Maintain a minimum of 15 percent residue cover immediately after planting where the purpose is erosion reduction.
Residue cover shall be measured by the line transect method or another approved procedure, and recorded.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

Field Office Technical Guide, Section IV.
Virginia Agricultural BMP Manual, current edition.

CONSERVATION PRACTICE STANDARD

RIPARIAN FOREST BUFFER (391)

Code 391 Unit: Ac.

DEFINITION

An area predominantly of trees and shrubs located adjacent to and up-gradient from watercourses or water bodies.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Create shade to lower water temperature and improve habitat.
- Provide a source of detritus and large woody debris.
- Reduce excess amounts of sediment, organic material, nutrients and
- pesticides in surface runoff and reduce excess nutrients and chemicals
- in shallow ground water flow.
- Restore riparian plant communities.
- Increase carbon storage.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

RIPARIAN FOREST BUFFER (391) - criteria

CRITERIA

The buffer shall consist of two zones. Zone 1 begins at the top of the channel bank and extends a minimum of 15 feet. Zone 2 begins at the edge of zone 1 and extends to the outer limit of the buffer.
Minimum total width shall be 35 feet. Where the purpose includes nutrient and sediment reduction, a width of 100 feet is recommended.
Species shall be native and adapted to the site hydrology.
Livestock shall be excluded from zone 1 at all times.
Where the buffer is established on cropland, the preconversion land use shall be documented.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

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CONSERVATION PRACTICE STANDARD

PRESCRIBED GRAZING (528)

Code 528 Unit: Ac.

DEFINITION

Managing the harvest of vegetation with grazing or browsing animals.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Improve or maintain desired species composition, structure and vigor.
- Improve or maintain quantity and quality of forage for grazing.
- Improve or maintain surface and subsurface water quality and quantity.
- Reduce soil erosion and maintain or improve soil condition.
- Manage fine fuel loads to achieve desired conditions.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

PRESCRIBED GRAZING (528) - criteria

CRITERIA

The quantity of forage available shall be adequate to meet the demand of the grazing animals while leaving the planned residual.
Grazing periods and rest periods shall be based on the rate of plant growth and the recovery requirement of the key species.
A minimum residual height shall be established for each key species and shall not be exceeded.
Livestock distribution shall be managed through water placement, fencing, salt and mineral placement, or herding.
Where this practice is applied to protect water quality, livestock access to the watercourse shall be controlled.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

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CONSERVATION PRACTICE STANDARD

FENCE (382)

Code 382 Unit: Ft.

DEFINITION

A constructed barrier to animals or people.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Facilitate the accomplishment of conservation objectives by providing a
- means to control movement of animals and people, including vehicles.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

FENCE (382) - criteria

CRITERIA

Fences shall meet the requirements of applicable local and state law.
Design shall be appropriate to the kind and class of livestock, the terrain and the intended management.
Where the purpose is livestock exclusion from a watercourse, the fence shall be located outside the area to be protected.
Gates and crossings shall be provided where access is required.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

Field Office Technical Guide, Section IV.
Virginia Agricultural BMP Manual, current edition.

CONSERVATION PRACTICE STANDARD

WATERING FACILITY (614)

Code 614 Unit: No.

DEFINITION

A permanent or portable device to provide an adequate amount and quality of drinking water for livestock or wildlife.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Provide access to drinking water for livestock or wildlife.
- Improve animal distribution.
- Protect streams, ponds and water supplies from contamination.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

WATERING FACILITY (614) - criteria

CRITERIA

Capacity shall be adequate for the number and kind of animals served.
The facility shall be located to improve grazing distribution and to reduce concentration near sensitive areas.
Footing or a heavy use area shall be provided where traffic will cause soil degradation.
Overflow shall be managed to prevent erosion.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
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REFERENCES

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CONSERVATION PRACTICE STANDARD

PIPELINE (516)

Code 516 Unit: Ft.

DEFINITION

A pipeline and appurtenances installed to convey water for livestock, wildlife or other agricultural uses.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Convey water to the point of use.
- Reduce erosion and improve water distribution.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

PIPELINE (516) - criteria

CRITERIA

Capacity shall be sufficient to deliver the required flow at the required pressure.
Depth of cover shall be sufficient to prevent freezing and mechanical damage.
Pressure relief, drainage and air release shall be provided as the design requires.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

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CONSERVATION PRACTICE STANDARD

HEAVY USE AREA PROTECTION (561)

Code 561 Unit: Ac.

DEFINITION

The stabilization of areas frequently and intensively used by people, animals or vehicles by establishing vegetative cover, by surfacing with suitable materials, or by installing needed structures.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Provide a stable, non-eroding surface.
- Protect water quality by reducing contaminant transport.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

HEAVY USE AREA PROTECTION (561) - criteria

CRITERIA

The area shall be graded to divert clean water away and to collect and convey contaminated runoff to a treatment component.
Surfacing material shall be of a thickness and gradation suited to the expected loading.
Roof runoff shall be collected and conveyed away from the area where a roofed structure is adjacent.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
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REFERENCES

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CONSERVATION PRACTICE STANDARD

CONSERVATION CROP ROTATION (328)

Code 328 Unit: Ac.

DEFINITION

A planned sequence of crops grown on the same ground over a period of time.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Reduce sheet, rill and wind erosion.
- Maintain or increase soil health and organic matter content.
- Reduce water quality degradation due to excess nutrients.
- Improve soil moisture efficiency.
- Reduce the concentration of salts and other chemicals.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

CONSERVATION CROP ROTATION (328) - criteria

CRITERIA

The rotation shall include crops that produce sufficient residue to meet the erosion objective.
The sequence shall be planned for the full rotation cycle and documented.
Where the purpose is nutrient management, the rotation shall be consistent with the nutrient budget.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer. Review it periodically and revise as the operation changes.

REFERENCES

Field Office Technical Guide, Section IV.
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CONSERVATION PRACTICE STANDARD

NUTRIENT MANAGEMENT (590)

Code 590 Unit: Ac.

DEFINITION

Managing the amount, source, placement, form and timing of the application of nutrients and soil amendments.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Budget, supply and conserve nutrients for plant production.
- Minimize agricultural nonpoint source pollution of surface and ground
- water resources.
- Properly utilize manure or organic by-products as a plant nutrient
- source.
- Protect air quality.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

NUTRIENT MANAGEMENT (590) - criteria

CRITERIA

A nutrient budget shall be developed for nitrogen, phosphorus and potassium.
Soil tests shall be no more than four years old at the time of plan development.
Application rates shall not exceed the budgeted amount for the planned yield.
Records of application shall be maintained for five years, including date, field, source, rate, method and the person applying.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

Field Office Technical Guide, Section IV.
Virginia Agricultural BMP Manual, current edition.

CONSERVATION PRACTICE STANDARD

TREE/SHRUB ESTABLISHMENT (612)

Code 612 Unit: Ac.

DEFINITION

Establishing woody plants by planting seedlings or cuttings, by direct seeding, or through natural regeneration.

PURPOSE

This practice may be applied to support one or more of the following purposes:

- Establish woody plants for forest products.
- Establish or restore natural plant communities.
- Create or improve habitat.
- Provide shade and shelter for livestock.

CONDITIONS WHERE PRACTICE APPLIES

On land where the identified resource concern can be addressed by this practice and where the criteria below can be met.

CONSERVATION PRACTICE STANDARD

TREE/SHRUB ESTABLISHMENT (612) - criteria

CRITERIA

Species shall be adapted to the site and to the stated purpose.
Planting stock shall be of suitable size and quality.
Site preparation shall control competing vegetation.
Where the purpose is shade for livestock, protection from browsing shall be provided until the stock is established.
This practice alone does not constitute silvopasture establishment.

CONSIDERATIONS

Consider the effect on the water budget, on downstream flows, and on habitat. Consider the compatibility of the practice with the producer's equipment and management.

PLANS AND SPECIFICATIONS

Prepare plans and specifications that describe the requirements for applying the practice to achieve its intended purpose.

OPERATION AND MAINTENANCE

Prepare an operation and maintenance plan with the producer.
Review it periodically and revise as the operation changes.

REFERENCES

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Virginia Agricultural BMP Manual, current edition.

RESOURCE CONCERN ASSESSMENT

Whole farm

Y = concern identified P = partially treated N = not a concern

SOIL

Concern	Exists	Treated	Note
Sheet and rill erosion	Y	Y	RUSLE2 attached; within T on no-till fie
Wind erosion	N	-	Not a concern at this site
Concentrated flow erosion	Y	Y	Two ephemeral gullies, fields 14 and 51
Streambank erosion	Y	Y	~180 ft at livestock access points
Subsidence	N	-	
Compaction	Y	P	Headlands and the barnyard approach
Organic matter depletion	N	-	No-till since 2019; OM trending up

WATER

Concern	Exists	Treated	Note
Nutrients in surface water	Y	Y	Primary concern
Nutrients in ground water	Y	P	Karst influence on the Frederick soils
Pathogens and chemicals	Y	Y	Direct livestock access to Cub Run
Sediment in surface water	Y	Y	Unbuffered west bank
Ponding and flooding	N	-	
Inefficient irrigation	N	-	Not irrigated
Inadequate livestock water	Y	Y	Single watering point on 96 ac

PLANTS

Concern	Exists	Treated	Note
Productivity, health and vigor	Y	Y	Stand vigour fair, lower paddocks
Structure and composition	Y	Y	Continuous grazing
Pest pressure	N	-	
Wildfire hazard	N	-	

ANIMALS

Concern	Exists	Treated	Note
Terrestrial habitat	Y	P	Buffer establishment will improve
Aquatic habitat	Y	Y	Shade and temperature in Cub Run
Feed and forage imbalance	Y	Y	Late summer forage gap

ENERGY

Concern	Exists	Treated	Note
Equipment and facility efficiency	N	-	
Farming practice efficiency	N	-	No-till already in place

RUSLE2 SOIL LOSS SUMMARY

Cropped fields

Rusle Program Version: 2.6.10.1 Database: Virginia 2024

Field	Soil	Management	Loss	T	SCI	<T?
12	FaB	Corn silage / rye cover, no-till	4.1	5	0.42	Yes
14	GgA	Corn silage / rye cover, no-till	3.6	5	0.48	Yes
17	InA	Corn silage / rye cover, no-till	2.9	5	0.51	Yes
51	FaA	Corn / wheat / soybean, no-till	2.2	5	0.61	Yes
52	MnB	Corn / wheat / soybean, no-till	3.4	5	0.55	Yes
71	WeC	Corn silage, no-till	7.8	3	0.18	No
72	CnB	Corn silage, no-till	5.1	4	0.29	No

Loss is average annual sheet and rill soil loss in tons per acre.
T is the soil loss tolerance for the mapped soil.
SCI is the Soil Conditioning Index. A positive index indicates soil organic matter is expected to increase under the management shown.

Fields 71 and 72 exceed T under the current management. The planned conservation crop rotation and cover crop are expected to bring both within tolerance. Recalculate after the first full rotation cycle.

ENVIRONMENTAL EVALUATION

NRCS-CPA-52 equivalent

This evaluation documents the review of the planned practices against applicable environmental requirements. It was completed by the planner on the basis of records review and site inspection.

FEDERALLY LISTED SPECIES

Finding: No effect

A review of the Virginia Fish and Wildlife Information Service and the USFWS IPaC system returned no federally listed species with suitable habitat within the project area. The planned practices are vegetative and do not involve in-stream work.

STATE LISTED SPECIES AND NATURAL HERITAGE RESOURCES

Finding: No conflict

A Natural Heritage Data Explorer search was run for the project boundary. No documented occurrences of state listed species or natural heritage resources fall within the project area or a 100 foot buffer.

WETLANDS

Finding: No conversion

Hydric inclusions are mapped within the Inman component on field 17. No practice in this plan involves drainage, filling or manipulation of a wetland. Buffer establishment on hydric soils is not a conversion.

FLOODPLAIN MANAGEMENT

Finding: Compliant

Portions of fields 21, 22 and 55 fall within the mapped 100 year floodplain. The planned practices are vegetative establishment and do not place structures or fill within the floodplain.

CULTURAL RESOURCES

Finding: No historic properties affected

A review of the Virginia Cultural Resource Information System returned no recorded archaeological sites or architectural resources within the area of potential effect. Ground disturbance is limited to planting, post setting and a buried waterline within previously disturbed agricultural ground.

PRIME AND UNIQUE FARMLAND

Finding: No conversion to non-agricultural use

Portions of the Frederick and Murrill components are prime farmland. The planned conversion is from cropland to perennial pasture, which remains an agricultural use.